THEDAS SILVA

SOFTWARE ENGINEER | AI ENGINEER

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PROFILE

AI-focused software engineer currently at Innodata Lanka, working in python on cross-platform tooling and on R&D with LLMs and RAG systems. Experienced in building scalable, production-ready solutions across AI, backend, automation, and full-stack systems, and I enjoy taking an idea from prototype to something that actually runs in production. Keen to work on AI systems full-time.

WORK EXPERIENCE

Innodata Lanka (Pvt) Ltd

Software Engineer | 2026 Jun – Present

- Develop Python-based backend applications, RESTful APIs, and AI-powered software solutions.
- Driving R&D on LLM and RAG systems, building intelligent application and workflow automation solution.
- Design automation pipelines for data extraction, transformation, and process optimization.
- Collaborate in Agile teams using Git, CI/CD, and code reviews to deliver scalable, production-ready software.

Junior Specialist, Innovation Lab - AI manage services | 2026 Jan - 2026 May (5 Months)

- Developed Python-based automation systems for web scraping, data extraction, and workflow automation.
- Designed ETL pipelines and dashboards for data processing, visualization, and analysis.
- Built reliable data workflows with validation, testing, debugging, and performance improvements.
- Integrated processed data into internal platforms to support business operations.

Web Scraping Freelance Team Lead | 2025 Jul - 2025 Dec (5 Months)

- Led a freelance engineering team delivering Python-based web scraping and automation solutions
- Managed project planning, task allocation, technical guidance, and delivery timelines.
- Developed scalable scraping frameworks and automation tools for structured data extraction.
- Conducted code reviews, mentored developers, and collaborated with clients to deliver tailored solutions.

Web Automation Developer | 2024 Oct - 2025 Jul (9 Months)

- Developed Python-based web automation and data extraction solutions for dynamic websites.
- Built Selenium and Playwright workflows for authentication, pagination, and dynamic content handling.
- Designed ETL pipelines and integrated APIs for data extraction, transformation, and secure exchange.
- Optimized automation with logging, error handling, scheduling, and monitoring for scalability and reliability.

Production Associate | 2024 Apr - 2024 Oct (6 Months)

- Processed and validated structured and unstructured data from multiple sources.
- Analyzed and processed data using Microsoft Excel and converted XML/PDF data into structured formats.
- Supported engineering tool validation, testing, and quality assurance processes
- Maintained data accuracy through validation, quality checks, and process improvements.

EDUCATION

BSc (Hons) in Information Technology **2023 - 2026**
Sri Lanka Insititute of Information Technology (SLIIT), Malabe

GCE Advanced Level (Commerce stream) | 2021 **2019 - 2021**
Ananda College, Colombo 10

PROJECTS

SmartERP – AI-Powered Enterprise ERP System (2026)

React | Tailwind CSS | Python | FastAPI | Qdrant | Ollama | MySQL | Redis | Azure Blob Storage

- Designed a high-performance async backend using FastAPI and SQLAlchemy 2.0, achieving sub-millisecond API responses for core operations.
- Engineered a scalable RAG pipeline using Qdrant and Ollama, enabling semantic search across enterprise records.
- Developed asynchronous Redis queue processing for AI inference and document embedding tasks, keeping the main API unblocked.
- Implemented RBAC and secure JWT authentication, extending data access policies to vector searches for multi-tenant data privacy.
- Abstracted storage and database interfaces with Alembic and object storage protocols, supporting local and Azure Blob environments.

Embase Conference AI Article Extraction System (2026)

Python | GPT APIs | PyMuPDF | pdfplumber | PyPDF2 | Pandas

- Developed an AI-powered pipeline to extract conference abstracts and structured metadata from large PDF proceedings.
- Optimized LLM prompts for article boundary detection, metadata extraction, and structured output generation.
- Improved extraction accuracy from below 40% to over 90% through prompt engineering, validation, and PDF processing
- Improvements.
- Fixed duplicate article detection issues and improved handling of complex PDF layouts and inconsistent metadata.

Research Project – AI-Based Smart Retail Recommendation System for Apparel (2026 Jan – 2026 Oct)

Python | FastAPI | LangChain | HuggingFace | ChromaDB | Google Vertex AI | Scikit-Learn |Flutter.

- Engineered a scalable AI-driven retail backend utilizing Python and FastAPI to serve complex machine learning and generative AI workflows to a cross-platform mobile application.
- Designed and implemented a Retrieval-Augmented Generation (RAG) system using LangChain and ChromaDB to act as a conversational virtual stylist, indexing fabric metadata and customer reviews to provide personalized fit recommendations.
- Utilized HuggingFace embedding models (all-MiniLM) to convert text data into dense vector representations, enabling high-accuracy semantic search within the RAG pipeline.
- Integrated Google Vertex AI (virtual-try-on-001) to build a photorealistic image-to-image virtual try-on pipeline, seamlessly compositing garments onto user photos via zero-shot generation.
- Deployed cascaded predictive models (Random Forest, XGBoost) to autonomously infer exact human body measurements and map cross-brand sizing, eliminating the need for manual customer input.

TECHNICAL SKILLS

- **Programming languages:** Python, SQL, Bash, javaScript, java
- **Frameworks and Tools:** NodeJS, React + Redux, FastAPI, PostgreSQL, TensorFlow, OpenCV, LLMs, LangChain, Generative AI, n8n, Git
- **Soft Skills:** Teamwork, Communication Skills, Management Skills, Quick Learner, Problem Solving

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ACHIVEMENTS & ACTIVITIES

- SLIIT Rugby Team - Member | 2023
- SLIIT Inter Faculty Sports Meet (Tag Rugby) - Captain & Man of the match | 2023
- Aquinas College - English Level 3 | 2022
- Ananda College Prefect Team - Member | 2020 - 2021
- Ananda College Rugby Team - Vice Captain | 2013 – 2021

CERTIFICATIONS

- Scrape the web with python | Le Wagon
- AI/ML Engineer - stage 1 | SLIIT

REFEREES

Hashan Rajitha Welihinda,
Senior Tech Lead,
Axiata Digital Labs,
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Yasodhya Perera,
Specialist - Software engineer,
London Stock Exchange Group (LSEG),
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